

PYP Attempt 2

1a. 18.56

$$\begin{array}{r} 2 \overline{) 18} \\ 2 \overline{) 9} 0 \\ 2 \overline{) 4} 1 \\ 2 \overline{) 2} 0 \\ 2 \overline{) 1} 0 \end{array}$$

$$18 \rightarrow 010010$$

$$\cdot 56 \rightarrow 100$$

$\therefore$ Ans: 010010.100 with round off error of 0.06

b. -7.5

$$\begin{array}{r} 2 \overline{) 7} \\ 2 \overline{) 3} 1 \\ 2 \overline{) 1} 1 \end{array}$$

$$7 \rightarrow 00111$$

$$\cdot 5 \rightarrow 100$$

$$000111.100 \quad \text{-ve sign so flip}$$

$$111000.011$$

$$\begin{array}{r} + \quad \quad \quad 1 \\ \hline 111000.100 \end{array}$$

$$\text{Ans: } 111000.100$$

c. 22.87

$$\begin{array}{r} 2 \overline{) 22} \\ 2 \overline{) 11} 0 \\ 2 \overline{) 5} 1 \\ 2 \overline{) 2} 1 \\ 2 \overline{) 1} 0 \end{array}$$

$$22 \rightarrow 10110$$

$$\cdot 87 \rightarrow \cdot 5 + \cdot 25 = 0.75 \text{ round off error of } 0.12$$

$\therefore$ Ans: 010110.110 round off error 0.12

d. 36.5

$$\begin{array}{r} 2 \overline{) 36} \\ 2 \overline{) 18} 0 \\ 2 \overline{) 9} 0 \\ 2 \overline{) 4} 1 \\ 2 \overline{) 2} 0 \\ 2 \overline{) 1} 0 \end{array}$$

$$36 \Rightarrow \underline{100100}$$

exceeds limit

Ans: not possible to express 36 within the given range.

2. $S_0 = 100$

$$\sigma = 15\% = 0.15$$

$$r = 5\% = 0.05$$

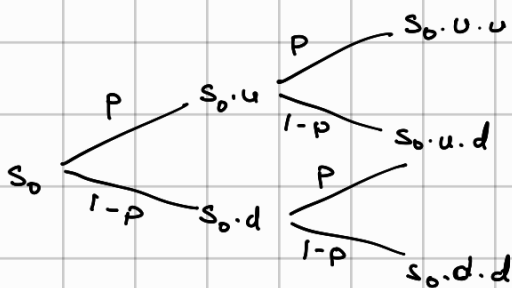
a. $\Delta t = 1$

$$u = e^{(r - \frac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t}} = e^{(0.05 - \frac{0.15^2}{2}) + 0.15} = 1.204439$$

$$d = e^{(r - \frac{\sigma^2}{2})\Delta t - \sigma\sqrt{\Delta t}} = e^{(0.05 - \frac{0.15^2}{2}) - 0.15} = 0.894415$$

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.05} - 0.894415}{1.204439 - 0.894415} = 0.500141$$

$$1 - p = 0.499859$$



$$\therefore S_{0,u} = 100 \times 1.204439 = 120.4439$$

$$S_{0,d} = 100 \times 0.894415 = 89.4415$$

$$S_{0,u,u} = 100 \times 1.204439^2 = 145.8633$$

$$S_{0,u,d} = 100 \times 1.204439 \times 0.894415 = 108.0582$$

$$S_{0,d,d} = 100 \times 0.894415^2 = 80.0515$$

b. Put option $(K - S_T)^+$; $K = 105$

At $T=2$

$$(K - S_{0,u,u})^+ = (105 - 145.8633)^+ = 0$$

$$(K - S_{0,u,d})^+ = (105 - 108.0582)^+ = 0$$

$$(K - S_{0,d,d})^+ = (105 - 80.0515)^+ = 24.9485$$

$$At \quad T=1$$

$$\begin{aligned} & (24.9485)(1-p)e^{-r\Delta t} \\ &= 24.9485 \times 0.499859 \times e^{-0.05} \\ &= 11.8625 \end{aligned}$$

$$\begin{aligned} \therefore V_0 &= 11.8625(1-p) \times e^{-0.05} \\ &= 11.8625 \times 0.49985 \times e^{-0.05} \\ &= 5.6404 \end{aligned}$$

$$3. \quad \begin{array}{l} x_1 = 0 \\ \hookrightarrow a = 0 \end{array} \quad \begin{array}{l} x_2 = 4 \\ \hookrightarrow b = 4 \end{array} \quad x^2 = x + 1 ; \quad \therefore f(x) = x^2 - x - 1$$

a. Bisection method 3 steps

$$\text{Let } tol = \frac{1}{10^4}$$

$$f(a) = 0^2 - 0 - 1 = -1$$

$$f(b) = 4^2 - 4 - 1 = 16 - 5 = 11$$

$$f(a)f(b) = -1 \times 11 = -11 < 0 \quad \text{fulfills condition}$$

$$\text{Step 1: } \frac{4-0}{2} = 2 > tol$$

$$\rightarrow c \leftarrow \frac{4+0}{2} = 2$$

$$\rightarrow f(c) = 2^2 - 2 - 1 = 4 - 3 = 1 \neq 0$$

$$\rightarrow f(a)f(c) = -1 \times 1 = -1 < 0$$

$$\therefore b \leftarrow c$$

Step 2:

$$\left(\frac{2-0}{2} \right) = 1 > tol$$

$$c \leftarrow \frac{2+0}{2} = 1$$

$$f(c) = 1^2 - 1 - 1 = 1 - 2 = -1 \neq 0$$

$$f(a)f(c) = -1 \times -1 = 1 > 0$$

$$\therefore a \leftarrow c$$

Step 3:

$$\left(\frac{2-1}{2}\right) = 0.5 > 2$$

$$c \leftarrow \frac{2+1}{2} = 1.5$$

$$f(c) = 1.5^2 - 1.5 - 1 = -0.25$$

$$f(a) = -1$$

$$f(a)f(c) = -1 \times -0.25 = 0.25 > 0$$

$$a \leftarrow c$$

Truncate here for this ques

4+ step 3: $a = 1.5$ $b = 2$

b. $x_1 = 0$ $x_2 = 4$ $tol = 10^{-4}$

Secant method 2 steps

$$f(x_2) = 4^2 - 4 - 1 = 11$$

$$f(x_1) = 0^2 - 0 - 1 = -1$$

Step 1:

$$|x_2 - x_1| = |4 - 0| = 4 > tol$$

$$\therefore x_3 = \frac{0f(x_2) - 4f(x_1)}{f(x_2) - f(x_1)}$$

$$= \frac{-4 \times -1}{11 - (-1)} = \frac{4}{12} = \frac{1}{3}$$

$$f\left(\frac{1}{3}\right) = \left(\frac{1}{3}\right)^2 - \frac{1}{3} - 1 = -\frac{11}{9}$$

$$x_1 \leftarrow x_2 \quad \therefore x_1 = 4$$

$$x_2 \leftarrow x_3 \quad \therefore x_2 = \frac{1}{3}$$

Step 2: $\left| \frac{1}{3} - 4 \right| = \frac{11}{3} > \text{tol}$

$$f(x_1) = f(4) = 11$$

$$f(x_2) = f\left(\frac{1}{3}\right) = -\frac{11}{9}$$

$$x_3 = \frac{4 \left(-\frac{11}{9}\right) - \left(\frac{1}{3}\right)(11)}{-\frac{11}{9} - 11} = \frac{4}{10}$$

$$f(x_3) = \left(\frac{4}{10}\right)^2 - \frac{4}{10} - 1 = -1.21 \neq 0$$

$$\therefore x_1 \leftarrow x_2 \quad \therefore x_1 = \frac{1}{3}$$

$$x_2 \leftarrow x_3 \quad \therefore x_2 = \frac{4}{10}$$

Truncate here for this question

$\therefore$ after 2 steps root between $\left[\frac{1}{3}, \frac{4}{10}\right]$

c. $x_1 = 0$ $x_2 = 4$ 2 steps false position
 $a = 0$ $b = 4$

$$f(a) = -1$$

$$f(b) = 11$$

$$f(a)f(b) = -1 \times 11 = -11 < 0$$

$$\text{tol} = 10^{-7}$$

Step 1:

$$b - a = 4 - 0 = 4 > \text{tol}$$

$$c = \frac{0 f(b) - 4 f(a)}{f(b) - f(a)} = \frac{-4 \times -1}{11 - (-1)} = \frac{4}{12} = \frac{1}{3}$$

$$f(c) = f\left(\frac{1}{3}\right) = -\frac{11}{9} \leftarrow \text{calculated in part b.}$$

$$\therefore f(a)f(c) = -1 \times -\frac{11}{9} = \frac{11}{9} > 0$$

$$\therefore a \leftarrow c \quad \therefore a = \frac{1}{3} \quad [b \text{ remains at } 4]$$

Step 2:

$$b - a = 4 - \frac{1}{3} = \frac{11}{3} > \text{tol.}$$

$$c = \frac{a f(b) - b f(a)}{f(b) - f(a)} = \frac{\frac{1}{3} \times (11) - 4 \left(-\frac{11}{9}\right)}{11 - \left(-\frac{11}{9}\right)} = \frac{4}{10}$$

$$f(c) = f\left(\frac{4}{10}\right) = -1.21 \quad \leftarrow \text{calculated in part b}$$

$$f\left(\frac{1}{3}\right) f\left(\frac{4}{10}\right) = -\frac{11}{9} \times -1.21 > 0$$

$$\therefore a \leftarrow c$$

$$\therefore a = \frac{4}{10} \quad b = 4$$

Truncate here for this ques.

$$\therefore \text{The root is between } \left[\frac{4}{10}, 4\right]$$

$$4. f(x) = \frac{1}{2} e^{-\frac{1}{2}x^2}$$

$$\int_0^2 f(x) dx = ?$$

$$a. E[f(x)] = \int_0^2 f(x) p(x) dx = \int_0^2 f(x) \frac{1}{2-0} dx = \frac{1}{2} \int_0^2 f(x) dx$$

$$\therefore 2 E[f(x)] = \int_0^2 f(x) dx$$

$$h = \int$$

∴ Pseudocode is

Algorithm: $\hat{\mu}_n = MC(n)$

∴ $S = 0$

for $i = 1$ to n do

Generate $\tilde{X}_i \sim U(0, 2)$

$h_i = h(\tilde{X}_i)$

$S += h_i$

end for

$\hat{\mu}_n = 2 \times (S/n)$

return $\hat{\mu}_n$

b. $g(x) = f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2$

$$f(0) = \frac{1}{2} e^{-\frac{1}{2} \cdot 0^2} = \frac{1}{2}$$

$$f'(x) = \frac{1}{2} e^{-\frac{1}{2}x^2} \cdot \frac{d(-\frac{1}{2}x^2)}{dx} = \frac{1}{2} e^{-\frac{1}{2}x^2} \cdot x \cdot -1 = -x \frac{1}{2} e^{-\frac{1}{2}x^2}$$

$$f'(0) = 0$$

$$\begin{aligned} f''(x) &= \frac{d(-x)}{dx} \times \frac{1}{2} e^{-\frac{1}{2}x^2} + \frac{d(\frac{1}{2} e^{-\frac{1}{2}x^2})}{dx} \cdot x \cdot -x \\ &= -\frac{1}{2} e^{-\frac{1}{2}x^2} + x^2 \frac{1}{2} e^{-\frac{1}{2}x^2} \end{aligned}$$

$$f''(0) = -\frac{1}{2}$$

$$\begin{aligned} g(x) &= \frac{1}{2} + 0(-x) + \frac{1}{2} \times -\frac{1}{2} \times x^2 \\ &= \frac{1}{2} - \frac{1}{4}x^2 \end{aligned}$$

$$\begin{aligned}
 \int_0^2 g(x) dx &= \int_0^2 \frac{1}{2} dx - \int_0^2 \frac{1}{4} x^2 dx \\
 &= \left[\frac{1}{2} x \right]_0^2 - \left[\frac{x^3}{12} \right]_0^2 \\
 &= \left(\frac{1(2)}{2} - 0 \right) - \left(\frac{2^3}{12} - 0 \right) \\
 &= 1 - \frac{8}{12} \\
 &= \frac{3-2}{3} = \frac{1}{3}
 \end{aligned}$$

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C. import numpy as np
import scipy.integrate as integrate

def f(x):
    fx = 0.5 * np.exp(-0.5 * (x**2))
    return fx

def g(x):
    gx = 0.5 - (0.25 * (x**2))
    return gx

def g_integral(g(x), a, b):
    g_int = integrate.quad(g(x), a, b)
    return g_int

beta = # defined earlier

def mcIntegrate(f, a, b, g, g_integral):
    s = 0
    N = 106 # define number of paths
    for i in range(1, N+1):
        x_i = np.random.uniform(a, b)

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$s_t = \int (x_i) + \text{beta}(g\text{-integral}(g(x_i), a, b) - g(x_i))$
 expectation = $2 * (S/N)$
 return expectation

$$s_a. \quad dV_t = \frac{\partial V}{\partial t} + (r-q)S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$$

$$\hat{V}_t = \frac{V_t}{B_t}$$

$$\frac{\partial \hat{V}}{\partial V} = \frac{1}{B_t}$$

$$\frac{\partial \hat{V}}{\partial B_t} = -\frac{V_t}{B_t^2}$$

$$\frac{\partial^2 \hat{V}}{\partial V^2} = 0$$

$$\frac{\partial^2 \hat{V}}{\partial V \partial B} = -\frac{1}{B^2}$$

$$\frac{\partial^2 \hat{V}}{\partial B_t^2} = \frac{2V_t}{B_t^3}$$

$$\begin{aligned}
 d\hat{V}_t &= \frac{1}{B_t} dV_t + \frac{-V_t}{B_t^2} dB_t + \frac{1}{2} \left(-\frac{1}{B_t^2} dV dB + \frac{2V_t}{B_t^3} (dB_t)^2 \right) \\
 &= \frac{1}{B_t} dV_t - \frac{V_t}{B_t^2} dB_t - \frac{1}{2 B_t^2} dV_t dB_t
 \end{aligned}$$

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (dS)^2$$

$$= \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} (r-q) dt + \frac{\partial V}{\partial S} \sigma dW_t + \frac{1}{2} \sigma^2 dt \frac{\partial^2 V}{\partial S^2}$$

$$d\hat{V}_t = \frac{1}{B_t} \left(\frac{\partial V}{\partial t} + \frac{\partial V}{\partial S} (r - q) + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma \frac{\partial V}{\partial S} dW_t$$

$$- \frac{V_t}{B_t} r dt - \frac{1}{2 B_t^2} \left(\sigma \frac{\partial V}{\partial S} dW_t \right) (r B dt) \xrightarrow{0}$$

$$d\hat{V}_t = \left(\frac{1}{B_t} \frac{\partial V}{\partial t} + (r - q) \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S^2} - r \frac{V}{B} \right) dt + \frac{\sigma}{B} \frac{\partial V}{\partial S} dW_t$$

$$\frac{\partial V}{\partial t} + (r - q) S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$$

$$\frac{\partial V}{\partial t} = rV - (r - q) S \frac{\partial V}{\partial S} - \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}$$

$$\therefore dV_t = \left(rV + (r - q)(1 - S) \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S^2} (1 - S^2) \right) dt + \frac{\sigma}{B} \frac{\partial V}{\partial S} dW_t$$

$$d\hat{V}_t = \frac{1}{B} \left(rV + (r - q) \frac{\partial V}{\partial S} (1 - S) + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial S^2} (1 - S^2) \right) dt + \sigma \frac{\partial V}{\partial S} dW_t$$

$$- \frac{rV}{B} dt$$

$$= \left(\frac{(r - q)}{B} \frac{\partial V}{\partial S} (1 - S) + \frac{\sigma^2}{2B} \frac{\partial^2 V}{\partial S^2} (1 - S^2) \right) dt + \frac{\sigma}{B} \frac{\partial V}{\partial S} dW_t$$

$$S_t = B_t e^x$$

$$\frac{\partial V}{\partial S} = \frac{\partial V}{\partial X} \frac{\partial X}{\partial S} = \frac{\partial (B\hat{V})}{\partial X} \frac{\partial V}{\partial S} = B_t \frac{\partial \hat{V}}{\partial X} \frac{\partial X}{\partial S} = \frac{B_t}{S_t} \frac{\partial \hat{V}}{\partial X} = \frac{B_t}{B_t e^x} \frac{\partial \hat{V}}{\partial X}$$

$$= \frac{1}{e^x} \frac{\partial \hat{V}}{\partial X}$$

$$\begin{aligned}
\frac{\partial^2 V}{\partial S^2} &= \frac{\partial}{\partial S} \left(\frac{\partial V}{\partial S} \right) \\
&= \frac{\partial}{\partial S} \left(\frac{\partial \hat{V}}{\partial X} \frac{1}{e^x} \right) \\
&= \frac{\partial}{\partial S} \left(\frac{\partial \hat{V}}{\partial X} \right) \frac{1}{e^x} + \frac{\partial \hat{V}}{\partial X} \frac{\partial}{\partial S} \left(\frac{1}{e^x} \right) \\
&= \frac{\partial}{\partial X} \left(\frac{\partial \hat{V}}{\partial X} \right) \frac{\partial X}{\partial S} \frac{1}{e^x} + \left(-e^{-x} \frac{\partial X}{\partial S} \right) \frac{\partial \hat{V}}{\partial X} \\
&= \frac{\partial}{\partial X} \left(\frac{1}{S} \frac{\partial \hat{V}}{\partial X} \right) - \frac{e^{-x}}{S} \frac{\partial \hat{V}}{\partial X} \\
&= \frac{1}{S e^{x_t}} \left(\frac{\partial^2 \hat{V}}{\partial X^2} - \frac{\partial \hat{V}}{\partial X} \right)
\end{aligned}$$

$$\begin{aligned}
&= \left(\frac{(r-q)}{B} \frac{\partial V}{\partial S} (1-S) + \frac{\sigma^2}{2B} \frac{\partial^2 V}{\partial S^2} (1-S^2) \right) dt + \frac{\sigma}{B} \frac{\partial V}{\partial S} dW_t \\
&= \left(\frac{(r-q)}{B} \frac{1}{e^{x_t}} \frac{\partial \hat{V}}{\partial X} (1-S) + \frac{\sigma^2}{2B} \frac{1}{S e^x} \left(\frac{\partial^2 \hat{V}}{\partial X^2} - \frac{\partial \hat{V}}{\partial X} \right) (1-S^2) \right) dt \\
&\quad + \frac{\sigma}{B} \frac{1}{e^x} \frac{\partial \hat{V}}{\partial X} dW_t
\end{aligned}$$

$$\begin{aligned}
&= \left(\frac{(r-q)}{B_t e^{x_t}} \frac{\partial \hat{V}}{\partial X} (1-B_t e^{x_t}) + \frac{\sigma^2}{2B_t^2 e^{2x_t}} \left(\frac{\partial^2 \hat{V}}{\partial X^2} - \frac{\partial \hat{V}}{\partial X} \right) (1-B_t^2 e^{2x_t}) \right) dt \\
&\quad + \frac{\sigma}{B_t e^{x_t}} \frac{\partial \hat{V}}{\partial X} dW_t
\end{aligned}$$

$$\therefore d\hat{V}_t = \frac{1}{B_t e^{x_t}} \left(\left((r-q) \frac{\partial \hat{V}}{\partial X} (1-B_t e^{x_t}) + \frac{\sigma^2}{2B_t e^{x_t}} \left(\frac{\partial^2 \hat{V}}{\partial X^2} - \frac{\partial \hat{V}}{\partial X} \right) (1-B_t^2 e^{2x_t}) \right) dt + \sigma \frac{\partial \hat{V}}{\partial X} dW_t \right)$$

$$\frac{\partial \hat{V}_t}{\partial t} - \frac{1}{B_t e^{x_t}} \left((r - q) \frac{\partial \hat{V}}{\partial x} (1 - B_t e^{x_t}) + \frac{\sigma^2}{2 B_t e^{x_t}} \left(\frac{\partial^2 \hat{V}}{\partial x^2} - \frac{\partial \hat{V}}{\partial x} \right) (1 - B_t^2 e^{2x_t}) \right) = 0$$

- b.
1. More stability: log transform in x_t reduces sensitivity to extreme prices. So more stable solutions.
 2. Reduced Variable Range: the log transform compresses the asset price range, so less computational domain & enhance efficiency.
 3. Simplified Boundary: Conditions for $x_t \pm \infty$ is easier to handle than $S_t \rightarrow 0$ or ∞ .
 4. Enhanced Grid Spacing: log transform is proportional to asset pricing changes.

$$c. \frac{\partial \hat{V}}{\partial t} \approx \frac{\hat{V}_{i,j+1} - \hat{V}_{i,j}}{\Delta t}$$

$$\frac{\partial \hat{V}}{\partial x} \approx \frac{\hat{V}_{i+1,j+1} - \hat{V}_{i-1,j+1}}{\Delta x}$$

$$\frac{\partial^2 \hat{V}}{\partial x^2} \approx \frac{\hat{V}_{i+1,j+1} - 2\hat{V}_{i,j+1} + \hat{V}_{i-1,j+1}}{(\Delta x)^2}$$

$$\Delta x = x_{i+1} - x_i$$

$$\therefore \frac{\hat{V}_{i,j+1} - \hat{V}_{i,j}}{\Delta t} = \frac{1}{B_t e^{x_t}} \left[(r - q) (1 - B_t e^{x_t}) \frac{\hat{V}_{i+1,j+1} - \hat{V}_{i-1,j+1}}{\Delta x} \right.$$

$$\left. + \frac{\sigma^2 (1 - B_t e^{x_t})}{2 B_t e^{x_t}} \left(\frac{\hat{V}_{i+1,j+1} - 2\hat{V}_{i,j+1} + \hat{V}_{i-1,j+1}}{(\Delta x)^2} - \frac{\hat{V}_{i+1,j+1} - \hat{V}_{i,j+1}}{\Delta x} \right) \right]$$

